TCS CORPORATE WEBSITE

Tab Block — Developer Guide

Role / audience	Development — Developers, technical leads, block owners, AI coding agents
Version	1.0
Status	Draft for review
Owner	Technical Lead / Block Owner (proposed)
Last validated	2026-11-13
Documents	1
Classification	INTERNAL — classification to be confirmed

Generated from the Markdown sources of docs/handover (branch migrated-page-content-edam); do not edit this PDF.

Contents

Tab Block — Developer Guide
- Summary
- Overview
- Source files
- Components, models and filters
- Fields
- Dependencies
- For Developers
- Extending the Block
- Responsive Behavior
- Accessibility
- Data Layer and Analytics
- Performance Considerations
- Security and Privacy
- For Architects
- Troubleshooting
- Known limitations of the block
- Change history of the block
- Reference pages
- Coverage of the required topics
- Screenshots
- Known limitations
- Open items
- Change history

Tab Block — Developer Guide

Attribute	Value
Document ID	DEV-B-tab
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	How the Tab block is built: files, model, structure, behaviour, variations, tests and limitations
Scope	blocks/tab/ and its components tab
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	13 November 2026
Last validated	2026-11-13 (model, files, imports from the code; reference pages from the live site)
Source files	blocks/tab/README.md, blocks/tab/_tab.json, blocks/tab/tab.css, blocks/tab/tab.js
Related documents	Variations, Test coverage, Authoring guide
Related AEM paths	—

1. Summary

Attribute	Value
Block	Tab
Folder	`blocks/tab`
Type	Child item
Category	child-item
Status	Active
Purpose	Starts a tab panel in a Tabs Container
Variations (summary)	Columns per row of its Column Breaks
Children	—
Code size	67 lines of JS and CSS

2. Overview

Business purpose. A simple, visible marker for "a new tab starts here", so tabbed content is built on the page with the standard components.

User experience. Visitors never see the Tab itself: its Title is the label of a tab button and its content is the panel. Authors see it in edit mode as a dashed chip with the Title.

Templates / page types. Any page that has a Tabs Container. **Data source:** its own fields.

Related blocks. Tabs (builds the tab list from the Tabs), Column Break (columns in a tab), Columns Container (the layout built in the tab). The Tabs block (`tabs-block`) has its own Tab items (`tabs-item`, with a Fragment), documented in the Tabs README; they aren't this block.

When to use. Inside a Tabs Container, once at the start of every tab.

When not to use. Outside a Tabs Container (it is offered only there; elsewhere it would show as a visible chip); in a Tabs block (use its Tab items).

Realistic use cases. Starting the tabs Consulting, Cloud and Cybersecurity of a services Tabs Container; naming a tab for links (`#cloud`); choosing the tab shown when the page opens; laying out a tab's content in two columns.

3. Source files

File	Link
`README.md`	GitHub
`_tab.json`	GitHub
`tab.css`	GitHub
`tab.js`	GitHub

4. Components, models and filters

Component	Title	Resource type	Template name	Model	Children allowed
`tab`	Tab	`core/franklin/components/block/v1/block`	Tab	`tab`	—

5. Fields

The Additional tab fields (anchor, hide, padding, margin, animation) are common to the blocks and described in the content modeling guide.

5.1 Tab (`tab`)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	Title	`title`	text	Yes	—	—	—
Content	ID	`tabId`	text	No	—	—	—
Content	Active tab	`activeTab`	boolean	No	—	—	—
Content	Columns per row	`classes_perrow`	select	No	`""`	5 options	—

"Required in model" is the model's `required` flag; whether a field is required in practice is stated in the authoring guide where it is known.

6. Dependencies

Import
`../../scripts/aem.js`
`../../utils/properties.js`

7. For Developers

```
blocks/tab/
├── README.md
├── _tab.json
├── tab.css
└── tab.js
```

Decorator/Utility	Purpose	Execution or dependency point
scripts/aem.js	Edge Delivery core library (block loading, CSS, metadata, optimized pictures)	tab.js: { toClassName }
utils/properties.js	readProperties, textOf, isOn, isOnByDefault (a toggle that defaults to on), optionOf, normalizeKey, textSource (key-value block properties)	tab.js: { isOn, readProperties, textOf }

Responsibilities. `tab.js` reads the Tab's fields (`readTab`, exported for `blocks/tabs/tabs.js`) and decorates the marker (label chip, `data-analytics="off"`, `tabs:tab-update`). `tab.css` styles the chip. The hiding on the page is in `blocks/tabs/tabs.css` (`.tabs > .tabs-panel > div:has(> .tab)`, shown again under `html.adobe-ue-edit`).

Input DOM. A key-value block (`div.tab`) with rows `title`, `tabid` / `tabId`, `activetab`; classes from the `classes_*` fields (`per-row-N`, `hideblock`). Without key rows, one row per field in dialog order (Title, ID, Active tab: the last cell of each row).

Rendered DOM.

```
<div class="tab block per-row-2" data-tab-title="Cloud" data-tab-id="cloud" data-tab-active="true" data-analytics="off">
  <p class="tab-title">Cloud</p>
</div>
```

Decoration flow. The Tabs block is loaded before the Tabs in its section, so `blocks/tabs/tabs.js` calls `readTab` on the undecorated Tab; `readTab` caches the values on `data-tab-title`, `data-tab-id`, `data-tab-active` so the later `decorate` (and re-reads) find them after the rows are gone. `decorate` then replaces the rows with `p.tab-title` and dispatches `tabs:tab-update` (bubbling, `detail`: `title`, `id`, `active`); the Tabs update the label of the tab whose panel contains the Tab.

Exports. `readTab(block)` → `{ title, id, active }`; `default decorate(block)`.

Parent / child contracts. The Tabs read `per-row-N` and `hideblock` from the Tab's classes (`groupContainerTabs`); a Tab with `hideblock` drops its group when not editing.

Error handling. Missing fields give an empty Title (the Tabs use "Tab {n}"), an empty ID and Active off.

8. Extending the Block

1. Add a field in `_tab.json`, run `npm run build:json` and `npm run lint`.
2. Read it in `readTab` (add its normalized key to `FIELDS` in dialog order and cache it on a `data-tab-*` attribute) and use it in `blocks/tabs/tabs.js` `groupContainerTabs`. A `classes_*` field arrives as a block class instead (as Columns per row).
3. Delivery rules: don't end field names in Text, Title, Type, Alt or MimeType; no sample value in the definition template.
4. Data layer: the Tab stays `data-analytics="off"`; describe any new data on the panel in `tabs.js`.
5. Backward compatibility: keep reading key-value and per-field rows.
6. No unit test covers the Tab; check the block library pages listed below. Update this README and run `npm run build:docs`; `npm run build:modules` after changing imports.

9. Responsive Behavior

The Tab's CSS has no media query: it is hidden on the page at every width and shown as a chip in edit mode at every width. The tab button and the columns built from its Column Breaks follow the Tabs and Columns Container rules.

Viewport	Layout	Image behavior	Interaction behavior
All widths	Hidden on the page; dashed chip in edit mode	None	None on the page; selectable in the editor

10. Accessibility

- The Tab adds no semantics of its own on the page: its Title becomes the accessible name of a `role="tab"` button, which labels its panel (`aria-labelledby`).
- The chip (`p.tab-title`) is shown only in edit mode; on the page its wrapper is hidden with `display: none`, so screen readers don't announce it.
- Hidden Tabs are removed from the tab list and the panels (not only visually hidden).

Author responsibilities: unique, descriptive Titles; a heading at the start of the tab's content; IDs that make sense in links.

Developer responsibilities: keep the Title as the tab button's text; keep the chip hidden outside edit mode (`blocks/tabs/tabs.css`); keep `data-analytics="off"`.

Keyboard behavior is the Tabs' (see the Tabs accessibility section); the Tab handles no keys.

11. Data Layer and Analytics

- The Tab is `data-analytics="off"`: it isn't a data-layer component and no click is measured on it.
- Its Title is the `dc:title` of its panel (`tcs-corp/components/tabs/item`, pushed by `blocks/tabs/tabs.js`) and of the `cmp:show` pushed when the tab is shown after the first one; its ID is the panel id used as the component id.
- The path Adobe Client Data Layer → Adobe Data Collection (Tags) → AEP Web SDK → Edge Network → Analytics / CJA is the intended one, but Tags and the Web SDK aren't configured yet (blocked: measurement plan). No consent mechanism is implemented.

Trigger	Data-layer event	Required properties	Analytics behavior
Page load	component state (by the Tabs)	panel id (the Tab's ID or `{base}-item-{n}`), `dc:title` (the Title)	The tab is listed as a panel
Its tab shown (not on load)	`cmp:show` (by the Tabs)	panel id, `dc:title`	Tab view

12. Performance Considerations

No network requests, observers or listeners; one small DOM replacement. The Tab is loaded as a block of its section (its CSS and JS are fetched once per page).

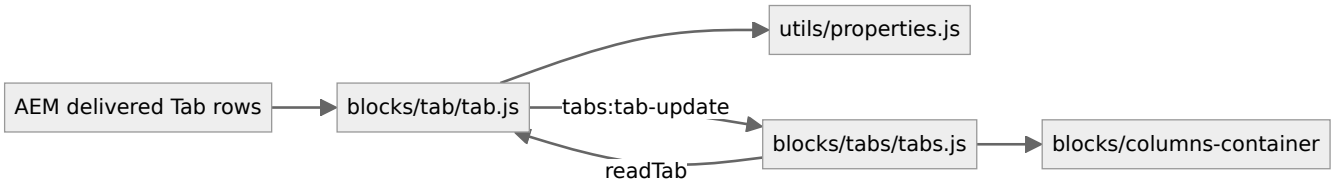
13. Security and Privacy

The Title is set with `textContent`; the ID is normalized with `toClassName`. No personal data.

14. For Architects

- **Role:** a marker block inside a section container: it delimits groups of sibling components, because Edge Delivery doesn't nest blocks in blocks.

- **Content model:** key-value block `tab`; its `classes_*` fields arrive as block classes (it is a block, not a section).
- **Rendering:** read by the Tabs before its own decoration; values cached in data attributes.
- **Data layer / Target / consent:** none of its own.
- **Performance / security:** negligible (no network, no listeners).



15. Troubleshooting

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Label "Tab 2" instead of the Title	Empty Title, or content before the first Tab	Select the Tab; check the tree order	Enter the Title; move stray components	Labels are the Titles
Chip visible on the page	The Tab isn't in a Tabs Container	Check the section for `tabs-section`	Move or delete the Tab	No chip
ID doesn't work in links	Normalized or de-duplicated id	Inspect the panel id	Use the exact ID	Link opens the tab
Active tab ignored	Another active Tab, a #hash, or the editor keeps the shown tab	Load without hash; check all Tabs	One Active tab; reload	Correct tab on load
ID, Active, Columns per row or Hide block change has no effect in the editor	Applied when the container is rendered again (R-C-7)	Reload the editor	Reload	Applied
Component in the wrong tab	Its position in the tree	Check the tree order	Move it after its Tab	Correct panel

Evidence to collect: page URL, the Tab's data attributes (`data-tab-*`) and classes, the panel ids.

16. Known limitations of the block

Limitation	Impact	Workaround	Planned correction
ID, Active tab, Columns per row and Hide block changes apply only when the Tabs Container is rendered again (re-audit R-C-7, deferred)	No immediate feedback in the editor	Reload the editor	Not recorded
A Tab outside a Tabs Container shows its chip on the page	Visible stray label	Keep Tabs in Tabs Containers (the filter offers it only there)	Not planned
Gap, alignment and breakpoint of the columns in a tab aren't authorable	Fixed medium gap, top, from 900px	Use a Columns Container section	Not planned

17. Change history of the block

Version	Change	Source	Date
Not recorded	README generated from verified implementation and approved handover documentation	Repository review	Not recorded
Handover 2.28	New Tab field Columns per row: a tab's Column Breaks lay its content out in columns	docs/handover revision history	20 October 2026
Handover 2.29	Block ecosystem re-audit: sample content removed from templates (re-audit R-C-10: the Tab no longer stores the title "Tab"; Tab IDs validated)	docs/handover revision history	20 October 2026

18. Reference pages

Kind	Page	Live URL
Curated design page	/design-block-master/en/curated-design-pages/story	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/curated-design-pages/story

19. Coverage of the required topics

Topic	Where
Purpose	Overview
Source files	generated: Files
Model, definition, filters, children	generated: Components
Dependencies	generated: Dependencies
Authored input / rendered output / DOM transformation	For Developers
Configuration fields, defaults, required fields	generated: Fields
Variations	TAB-VARIATIONS.md
Responsive behaviour	Responsive Behavior
Accessibility and keyboard	Accessibility
Analytics / data layer	Data Layer and Analytics
Performance	Performance Considerations
Security	Security and Privacy
Extending	Extending the Block
Known limitations	Known Limitations
Troubleshooting	Troubleshooting
Tests	TAB-TEST-COVERAGE.md
SEO behaviour	Not recorded
Caching considerations	Not recorded (Edge Delivery default caching; no block-specific rule)
Ownership	Technical Lead / Block Owner (proposed)

20. Screenshots

> **Screenshot required**
>
> - Required screen: Tab on its block library page
> - Required role/access: none (published page)

- Required navigation steps: Open https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/tab
- Reason unavailable: No block library page of this block was found under /design-block-master/en/block-library
- Owner: Content Lead / AEM Authoring Lead (proposed)
- Target completion date: To be confirmed

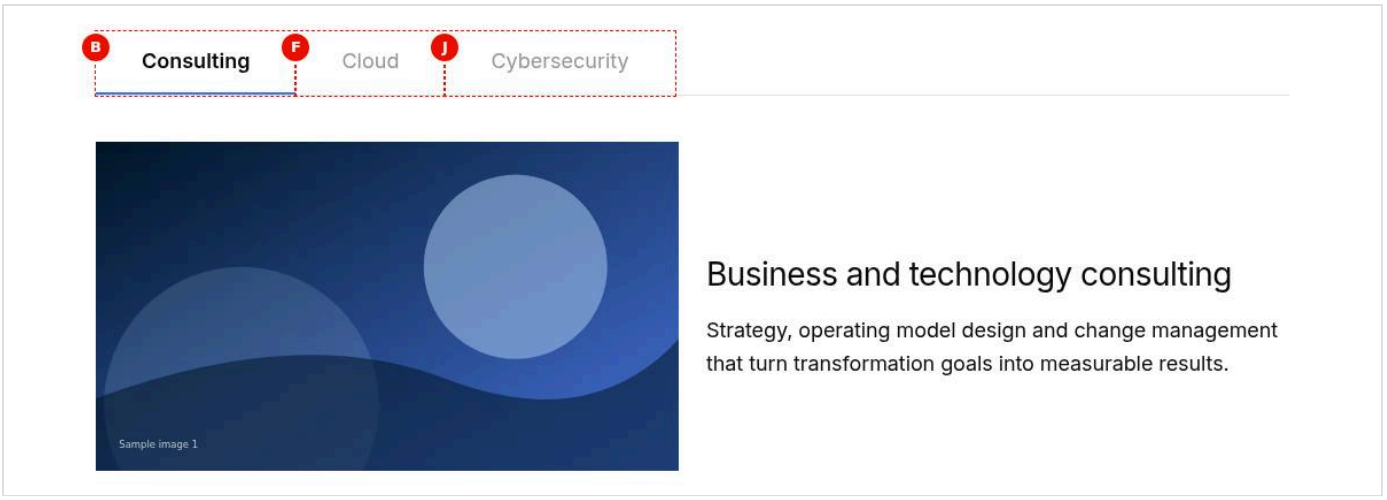


Figure: Desktop rendering — A Tabs Container at 1200 px on the page: the Tab markers' Titles appear only as the tab buttons Consulting (selected), Cloud and Cybersecurity; no marker is visible in the panel, which shows the Columns with photo-1.svg on the left and the H3 Business and technology consulting with its text on the right. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/tab-desktop-view.jpg) (handover screenshot; rendering at the time it was taken).

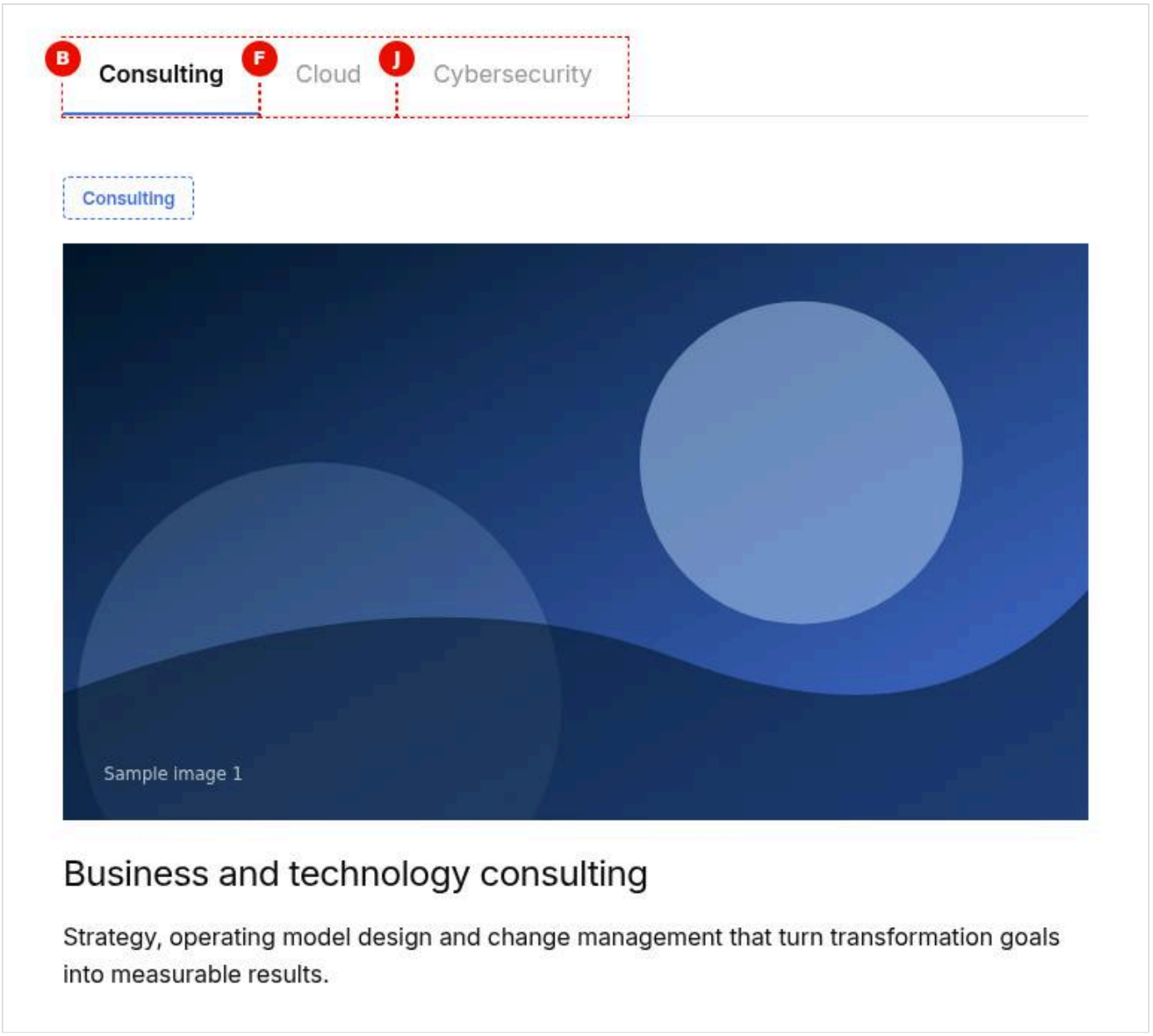


Figure: Tablet rendering — The same Tabs Container at 768 px in Universal Editor edit mode: the tab list Consulting (selected), Cloud and Cybersecurity, the dashed brand chip Consulting at the top of the panel, and below it the stacked Columns with photo-1.svg, the H3 Business and technology consulting and its text. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/tab-tablet-view.jpg) (handover screenshot; rendering at the time it was taken).

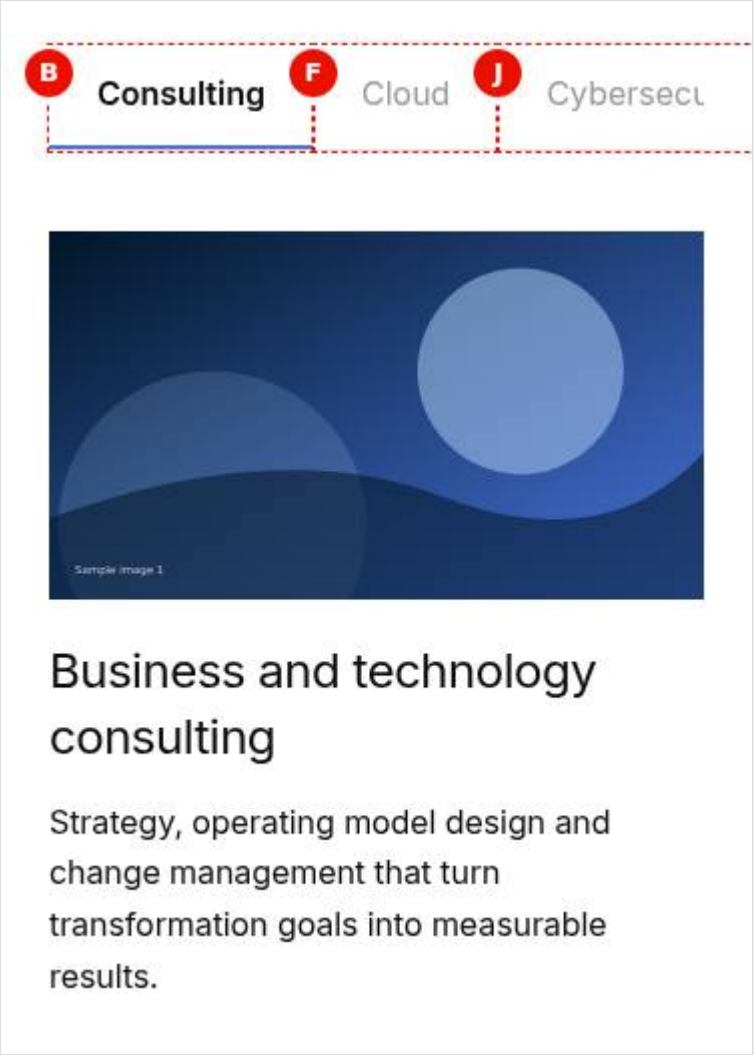


Figure: Mobile rendering — The Tabs Container at 375 px on the page: the tab list shows Consulting (selected), Cloud and a cut-off Cybersecu..., no marker in the panel; the Consulting panel is stacked: photo-1.svg, the H3 Business and technology consulting on two lines and its text. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/tab-mobile-view.jpg) (handover screenshot; rendering at the time it was taken).

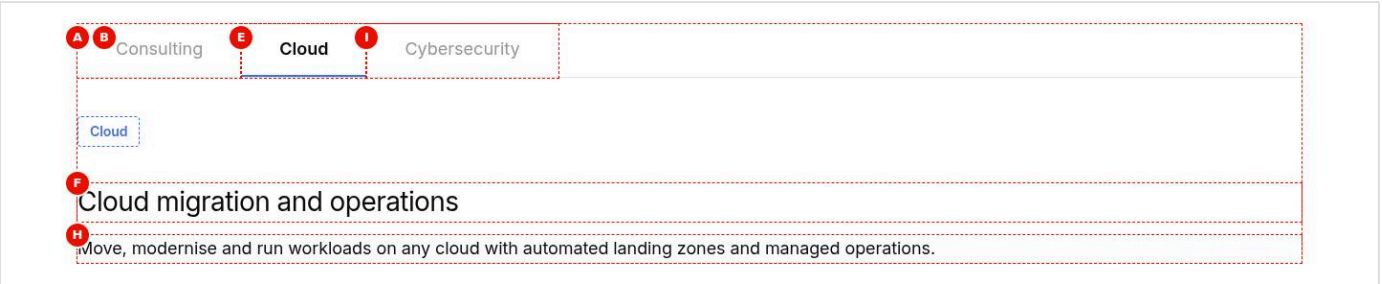


Figure: Standard Tab variation — Edit mode at 1440 px after selecting the Cloud tab button: Cloud is selected in the tab list Consulting, Cloud, Cybersecurity; the dashed chip Cloud sits at the top of the panel, followed by the H3 Cloud migration and operations and its text. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/tab-standard.jpg) (handover screenshot; rendering at the time it was taken).

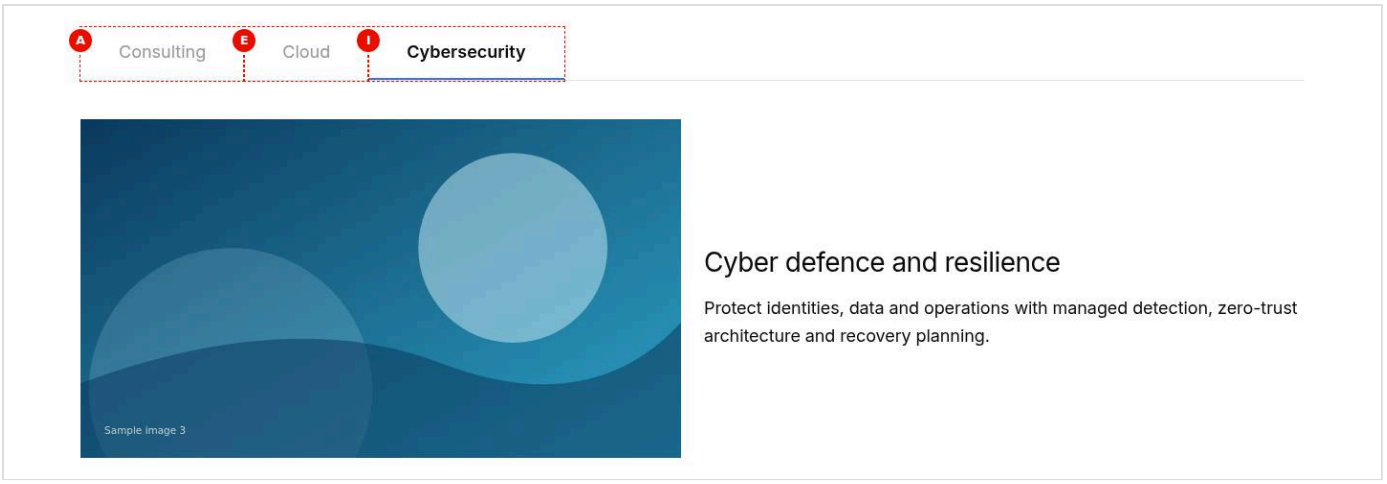


Figure: Active Tab variation — A Tabs Container at 1440 px loaded with its third tab, Cybersecurity, selected because that Tab has Active tab on; the panel shows the Columns with photo-3.svg on the left and the H3 Cyber defence and resilience with its text on the right. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/tab-active.jpg) (handover screenshot; rendering at the time it was taken).

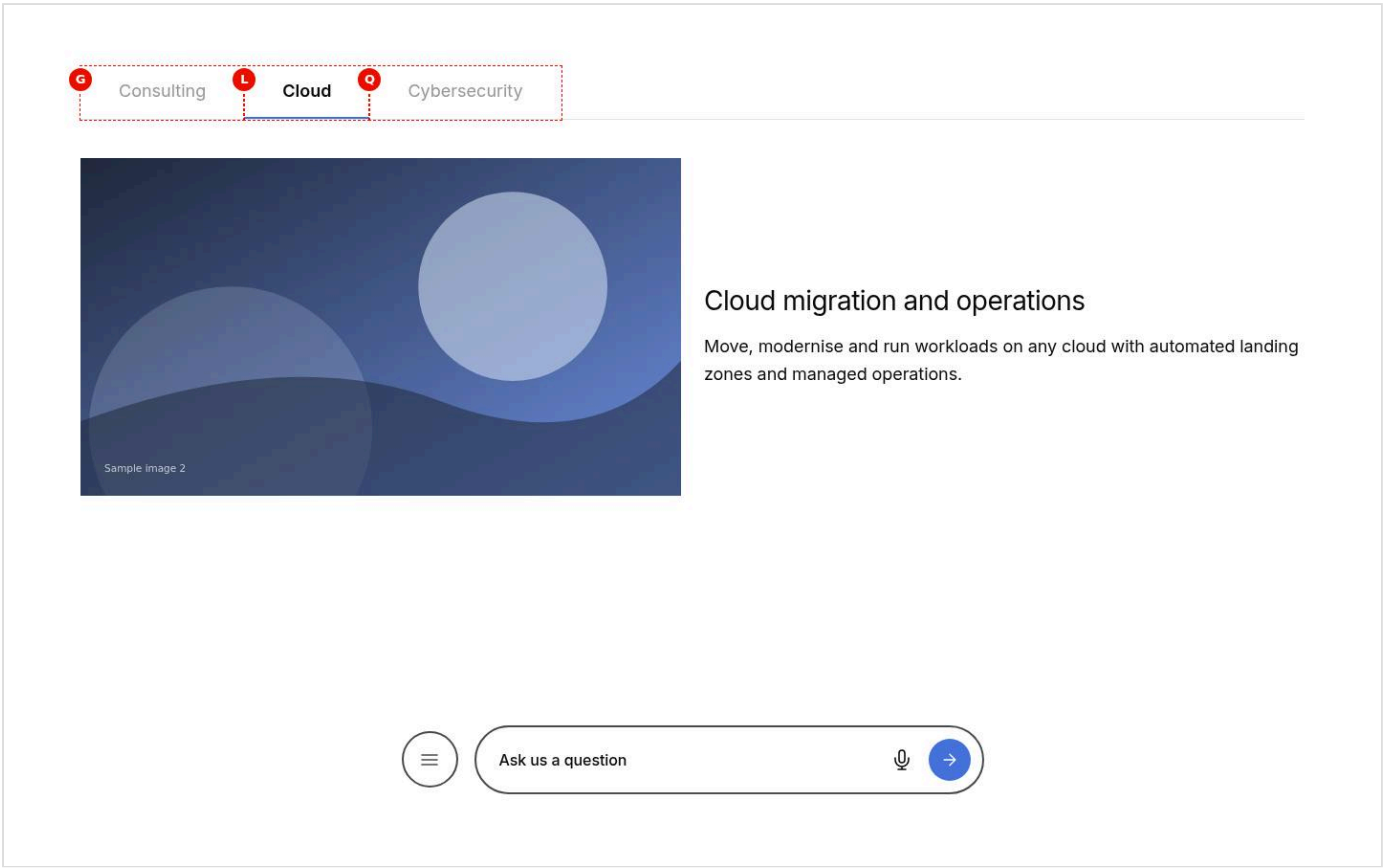


Figure: Tab with ID variation — The services page at 1440 px after #cloud is added to the address: the page has scrolled past the H1 and image so the Tabs sit just below the header area, with Cloud selected and its Columns (photo-2.svg, the H3 Cloud migration and operations and its text) shown; the footer's floating Ask us a question bar is at the bottom. The address bar itself is not in the screenshot. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/tab-deep-link.jpg) (handover screenshot; rendering at the time it was taken).

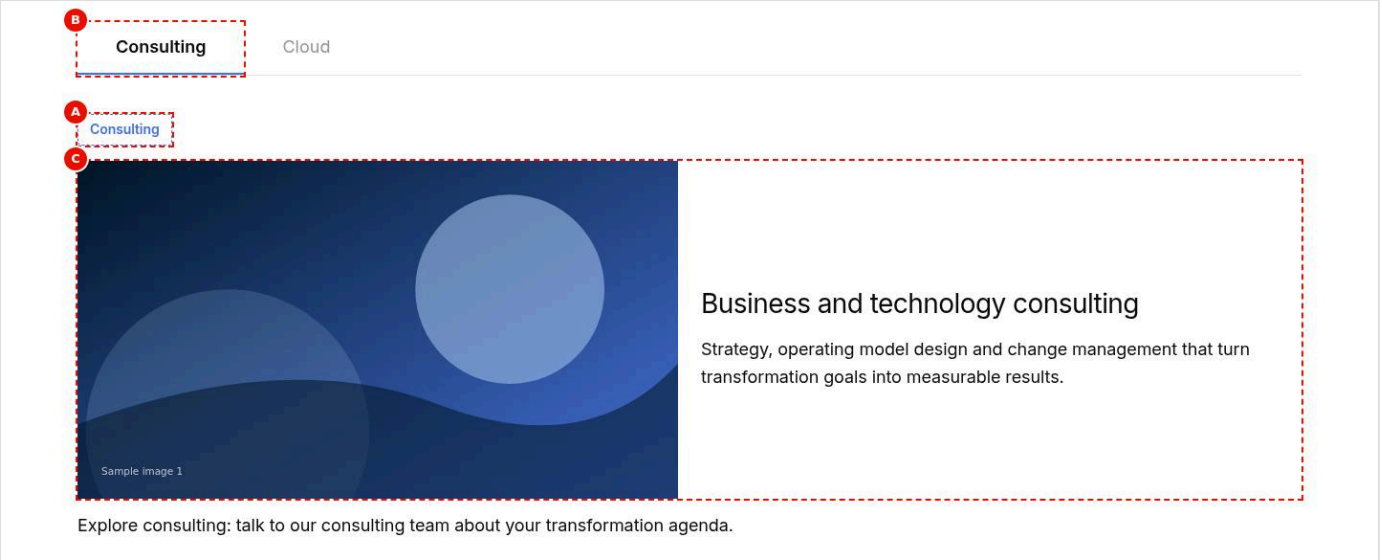


Figure: Universal Editor canvas — Edit mode preview at 1440 px with letters: A on the dashed Tab chip Consulting at the top of the panel, B on the selected tab button Consulting (Cloud next to it), C on the Columns with photo-1.svg and the H3; the Text Explore consulting: talk to our consulting team about your transformation agenda follows outside C. No content tree is shown. (Annotated preview in editor mode with sample content; letters as in the annotations list, not a Universal Editor screenshot; file screenshots/tab-authoring-view.jpg) (handover screenshot; rendering at the time it was taken).

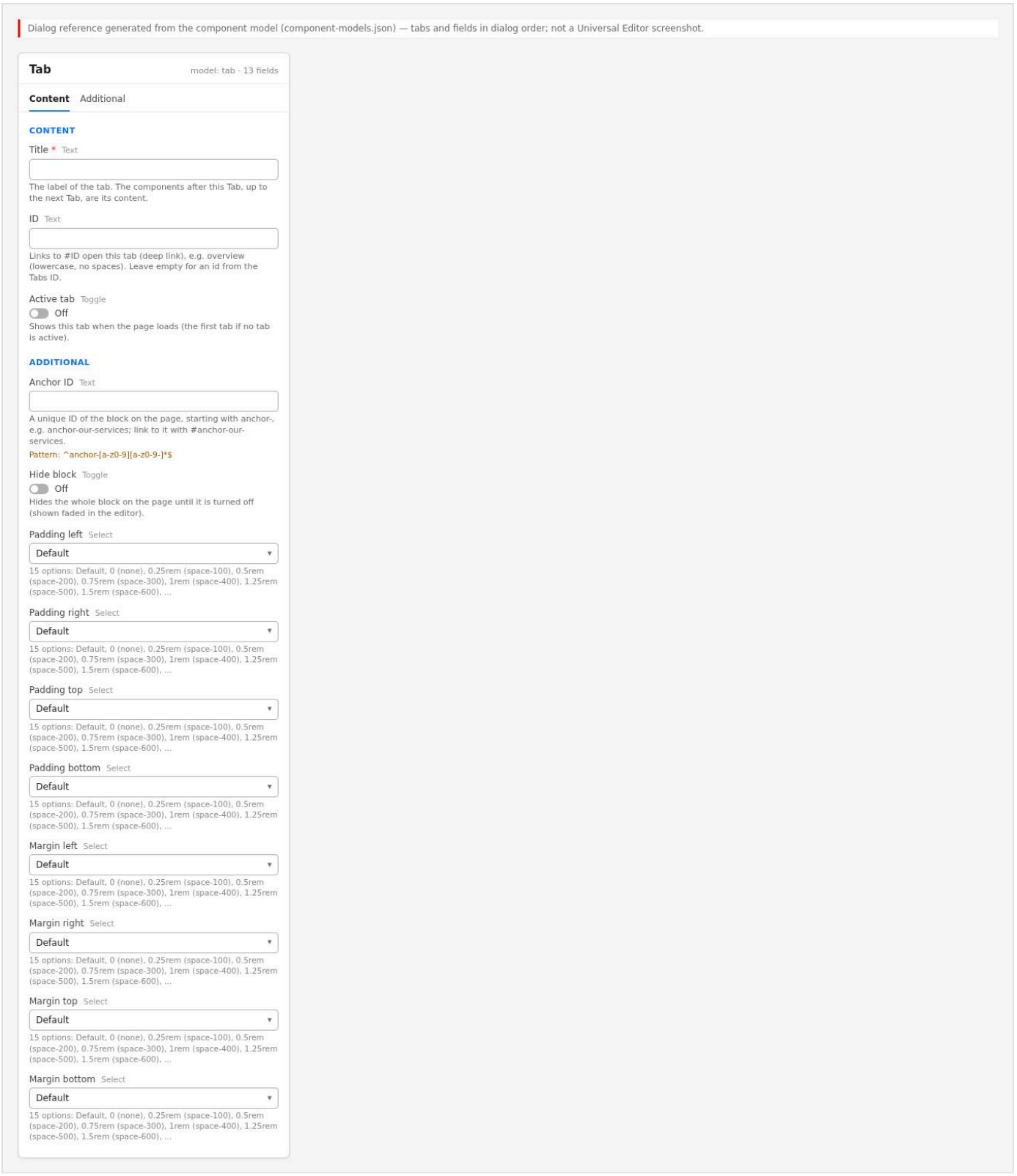


Figure: Properties panel — The properties panel of a Tab with the Content tab (Title, ID, Active tab) and the Additional tab. (Dialog reference generated from the component model, not a Universal Editor screenshot; file screenshots/tab-properties-panel.jpg) (handover screenshot; rendering at the time it was taken).

21. Known limitations

- Universal Editor dialogs were not captured: no author access in this program (see the access blockers).

22. Open items

- None.

23. Change history

Version	Date	Author	Change
1.0	13 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages